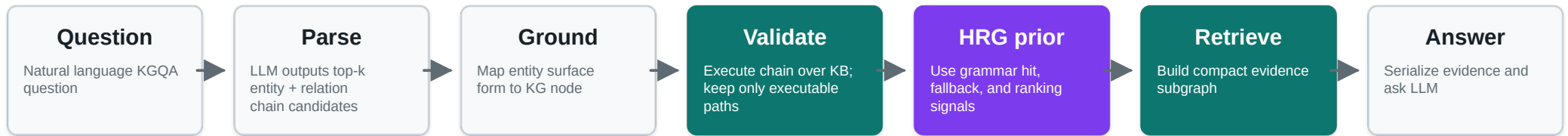


## HRG-guided KG-RAG: offline grammar prior and online evidence retrieval

Goal: extract structural priors from the KG, then use them during QA to select compact, executable evidence.



### What becomes explainable?

**selected\_entity**

topic node grounded in KG

**relation\_labels**

ordered KB relation labels

**HRG signal**

grammar hit / ranking prior

**spine\_edges**

printable evidence triples

**final\_context**

small context sent to LLM

### Main evidence claim

MetaQA: BFS 4415.83 context tokens

HRG-Proposed-triple: 77.26 context tokens

HRG is evaluated as a structural prior, not only compression

### Claim boundary

Not a claim of universal accuracy improvement

MLPQ / KQAPro remain hard cases

HRG is a structural prior for evidence retrieval

## Example retrieval: BFS neighborhood vs HRG-guided evidence

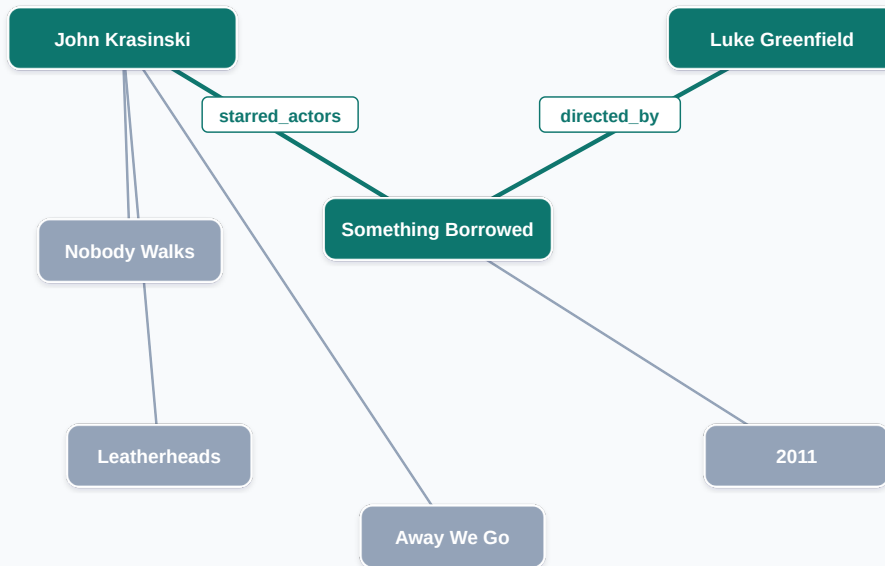
Question: which person directed the movies starred by [John Krasinski]

### Baseline-BFS context

Collects all nearby facts under BFS depth

86 edges

1092 ctx tokens

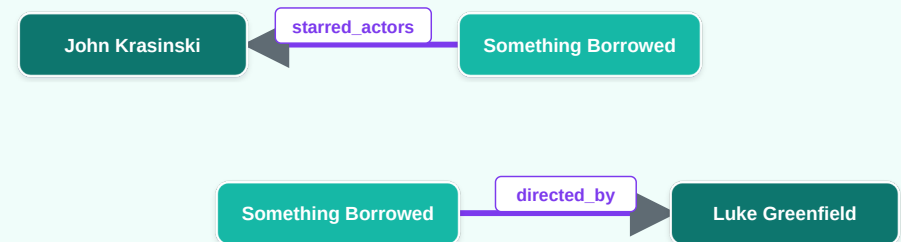


### HRG-Proposed context

Keeps compact evidence triples selected by KB + HRG signals

11 edges

257 ctx tokens



### Printable structure

selected\_entity = John Krasinski

relation labels = starred\_actors, directed\_by

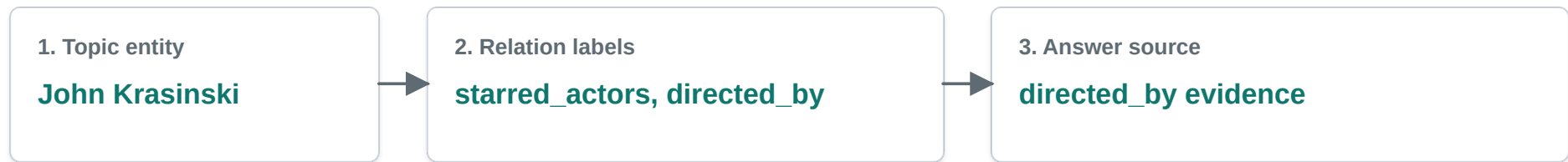
answers from directed\_by evidence

Same question, compact context, explicit KB triples

## What the explanation looks like for one question

Question

**which person directed the movies starred by [John Krasinski]**



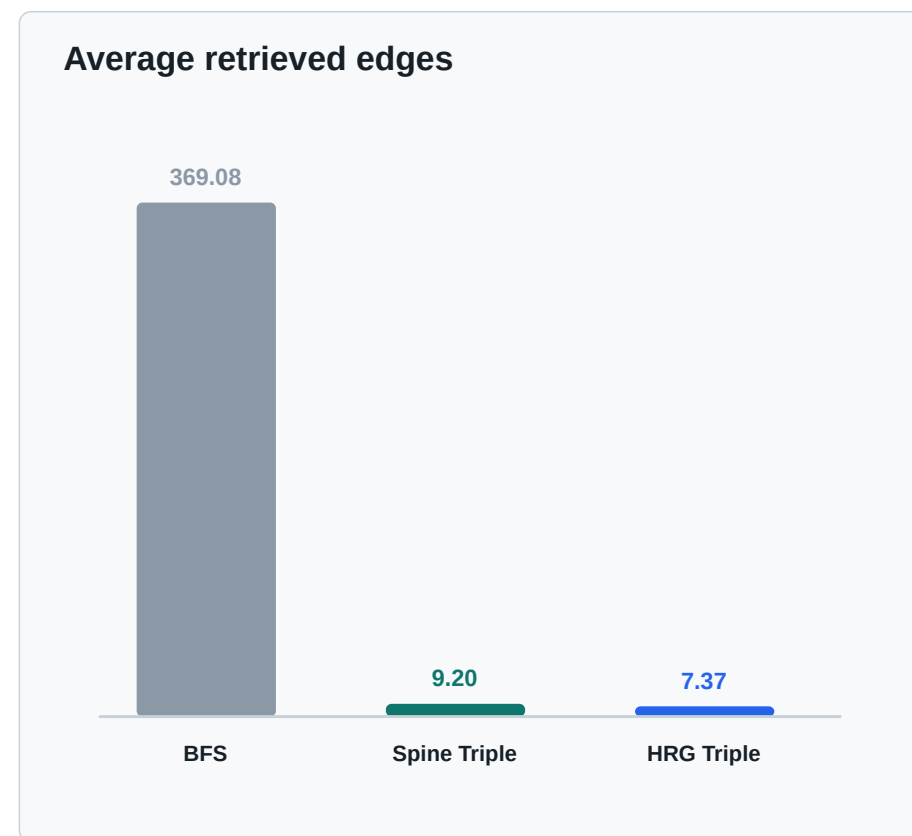
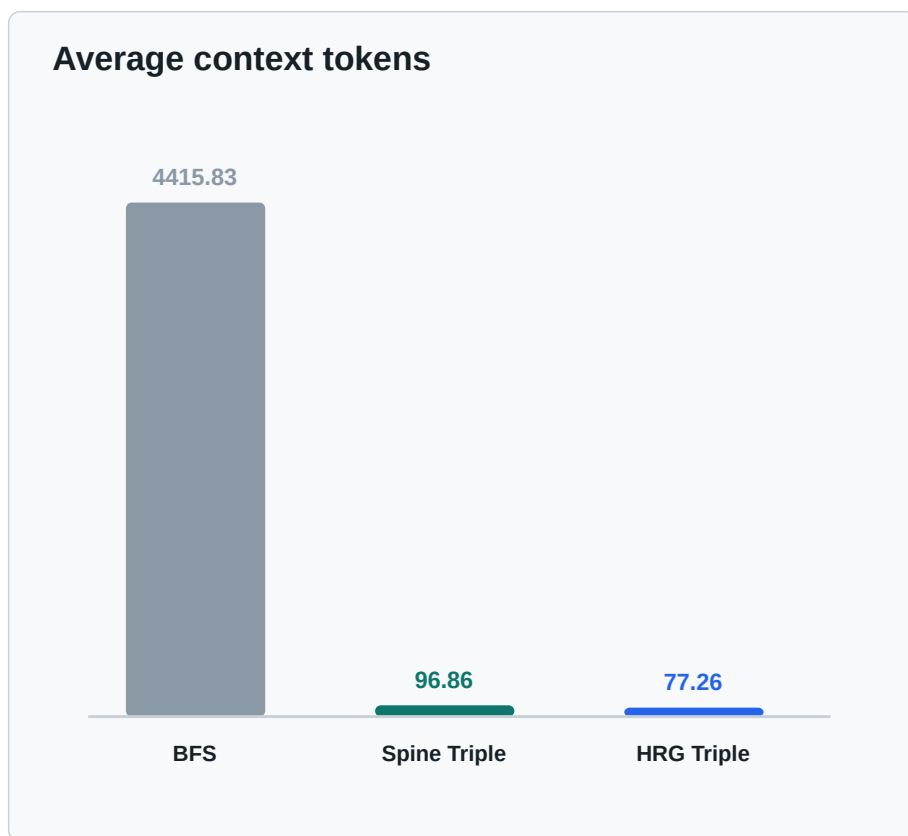
## KB evidence triples printed by the system

| Head               | Relation       | Tail            | Role                 |
|--------------------|----------------|-----------------|----------------------|
| Something Borrowed | starred_actors | John Krasinski  | matches topic entity |
| Something Borrowed | directed_by    | Luke Greenfield | one answer evidence  |

### Interpretability claim

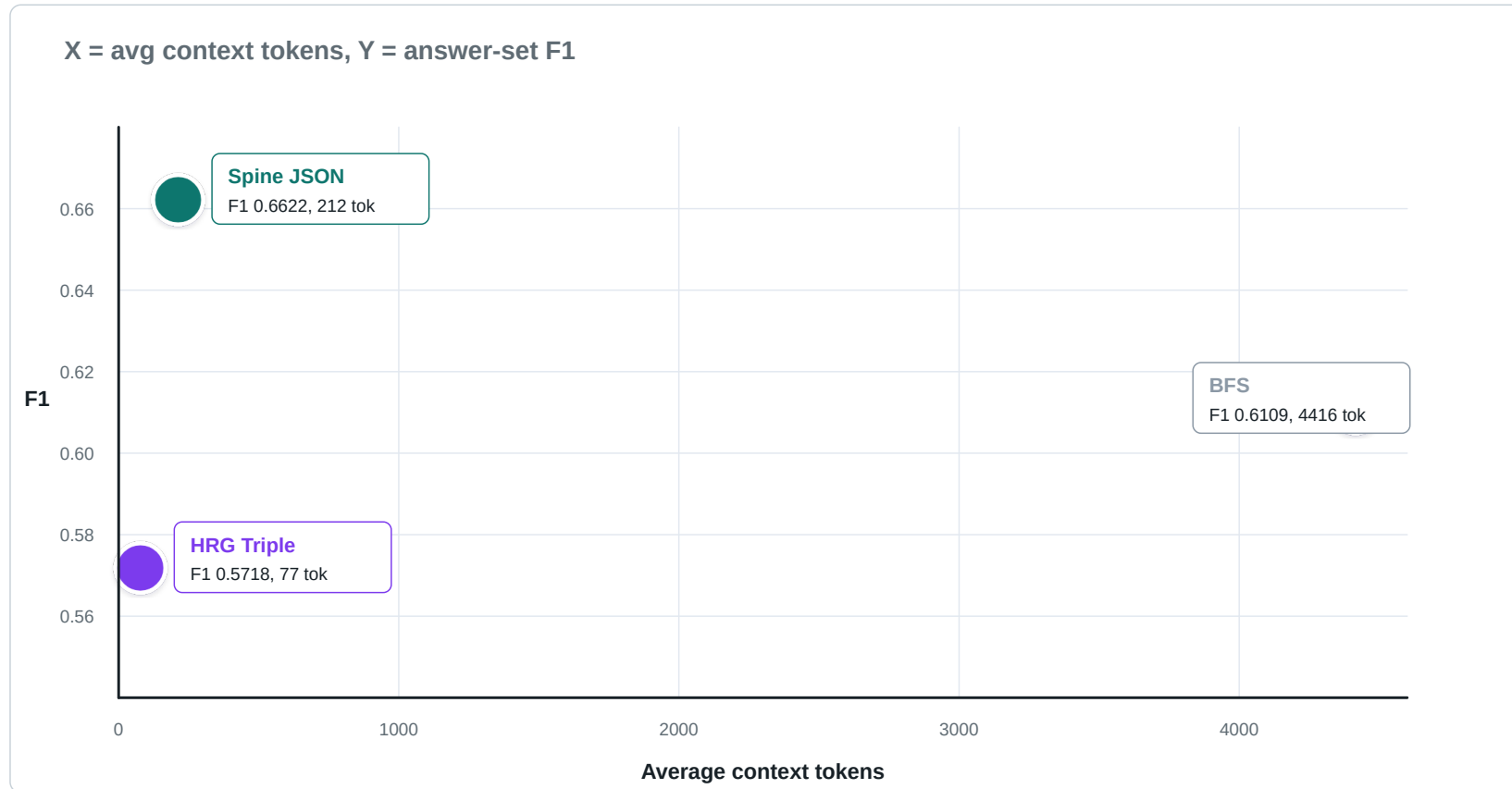
This explains retrieved KB evidence and HRG signals, not hidden LLM reasoning.

## MetaQA compression: HRG-Proposed context and subgraph size vs BFS



**HRG-Proposed-triple uses 1.75% of BFS context in this MetaQA setting.**

## MetaQA quality-cost trade-off: answer-set F1 vs context



### Takeaway

Main claim:  
quality-cost  
trade-off.

Report F1,  
evidence, and  
tokens together.

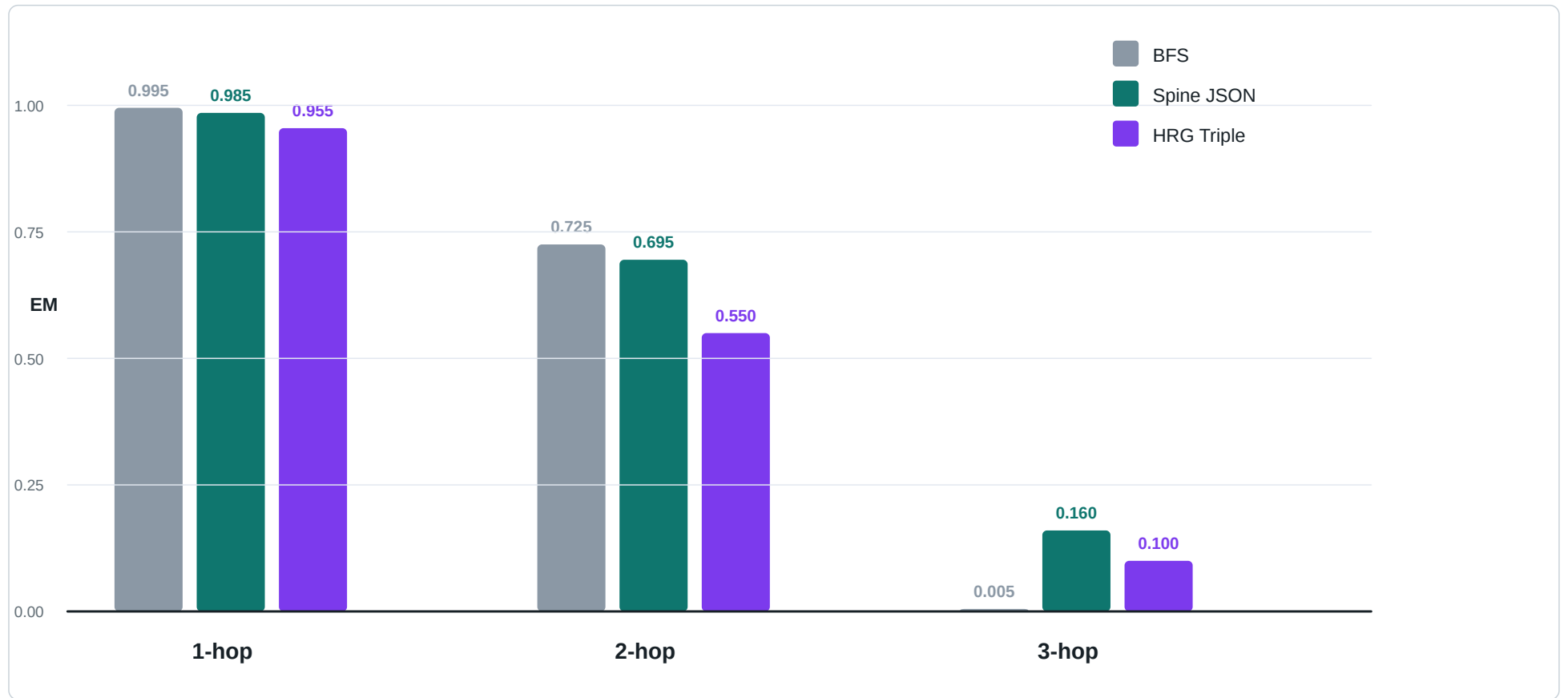
## Where HRG-Proposed evidence is strong or limited

| Dataset    | Answer quality | Token reduction    | How to present it         |
|------------|----------------|--------------------|---------------------------|
| MetaQA     | Usable         | Very strong        | Main compression evidence |
| WikiMovies | Near ceiling   | Small graph        | BFS already compact       |
| MLPQ       | Hard case      | HRG helps evidence | Cross-lingual relations   |
| KQAPro     | Hard case      | HRG helps evidence | Operators / qualifiers    |

### Recommended defense

Use MetaQA for compression; use MLPQ/KQAPro to show HRG evidence coverage under hard semantics.

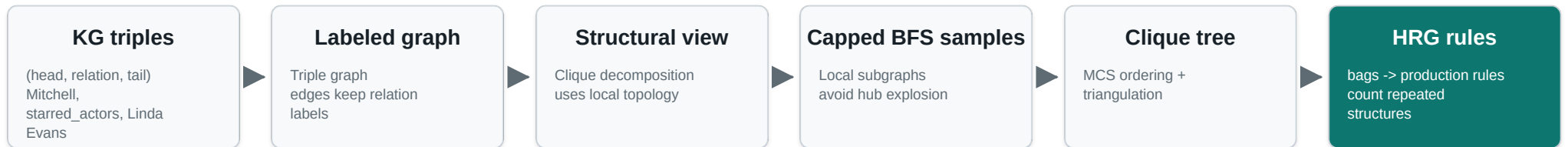
## MetaQA hop analysis: compare BFS, Spine, and HRG-Proposed



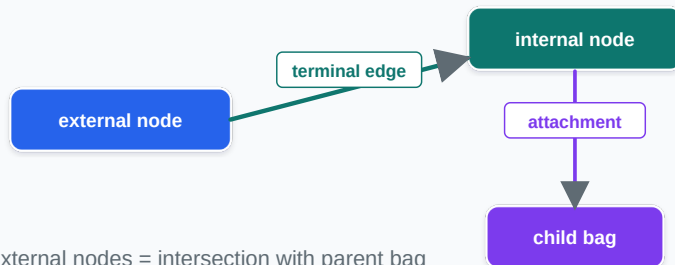
3-hop shows why relation-chain evidence must be evaluated by hop depth.

## Offline HRG grammar extraction

This figure explains how structural grammar rules are learned from the KG before question answering.



### How one bag becomes a rule



external nodes = intersection with parent bag  
internal nodes = new nodes introduced in this bag  
terminal edges = KG edges assigned to this bag

### Resulting production rule

LHS:  $N / \text{rank}$

RHS:

terminal\_edges = [(a, relation, b), ...]

nonterminals = [N / rank with attachments]

count = how often this structure appears

**Use this as a structural prior, not as a full grammar decoder.**

## KB relation-label validation with actual triples

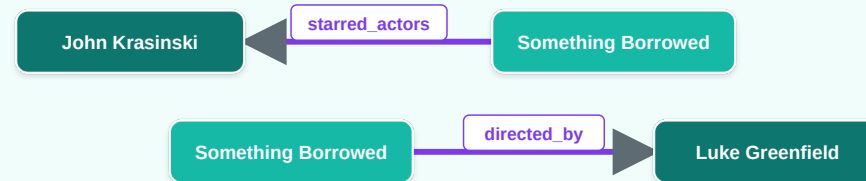
Question: which person directed the movies starred by [John Krasinski]?

LLM candidate

entity = John Krasinski  
relation labels =  
starred\_actors, directed\_by

validate

Validated KB evidence triples



### Validation rule

The relation label must appear in actual KB triples.

The topic entity and answer must be supported by retrieved evidence triples.

HRG contributes grammar-hit and ranking signals; it does not create new relation names.

valid evidence builds compact context

## Failure analysis: where HRG-Proposed still breaks

| Dataset                                                                              | ok  | no_candidates | no_valid_chain | Interpretation                                                   |
|--------------------------------------------------------------------------------------|-----|---------------|----------------|------------------------------------------------------------------|
| MetaQA                                                                               | 397 | 203           | 0              | High grammar-hit setting; compact evidence is usually available. |
|    |     |               |                | n=600                                                            |
| WikiMovies                                                                           | 196 | 4             | 0              | Mostly stable because questions are simple 1-hop.                |
|    |     |               |                | n=200                                                            |
| MLPQ                                                                                 | 389 | 0             | 0              | Cross-lingual grounding and relation alignment remain difficult. |
|    |     |               |                | n=389                                                            |
| KQAPro                                                                               | 272 | 317           | 9              | Operators and qualifiers exceed simple relation-chain evidence.  |
|  |     |               |                | n=598                                                            |

Use this figure to explain limits; HRG helps retrieval evidence but does not solve every semantic operator.

## Dataset semantics explain the results

MetaQA

**relation-chain friendly**

Question style

who directed films starred by [Linda Evans]?

Required structure

**relation labels: starred\_actors, directed\_by**

WikiMovies

**mostly 1-hop**

Question style

what films did Michelle Trachtenberg star in?

Required structure

**single KB relation label**

MLPQ

**cross-lingual path**

Question style

English question with en DBpedia / zh DBpedia evidence

Required structure

**cross-lingual entity and relation alignment**

KQAPro

**semantic program**

Question style

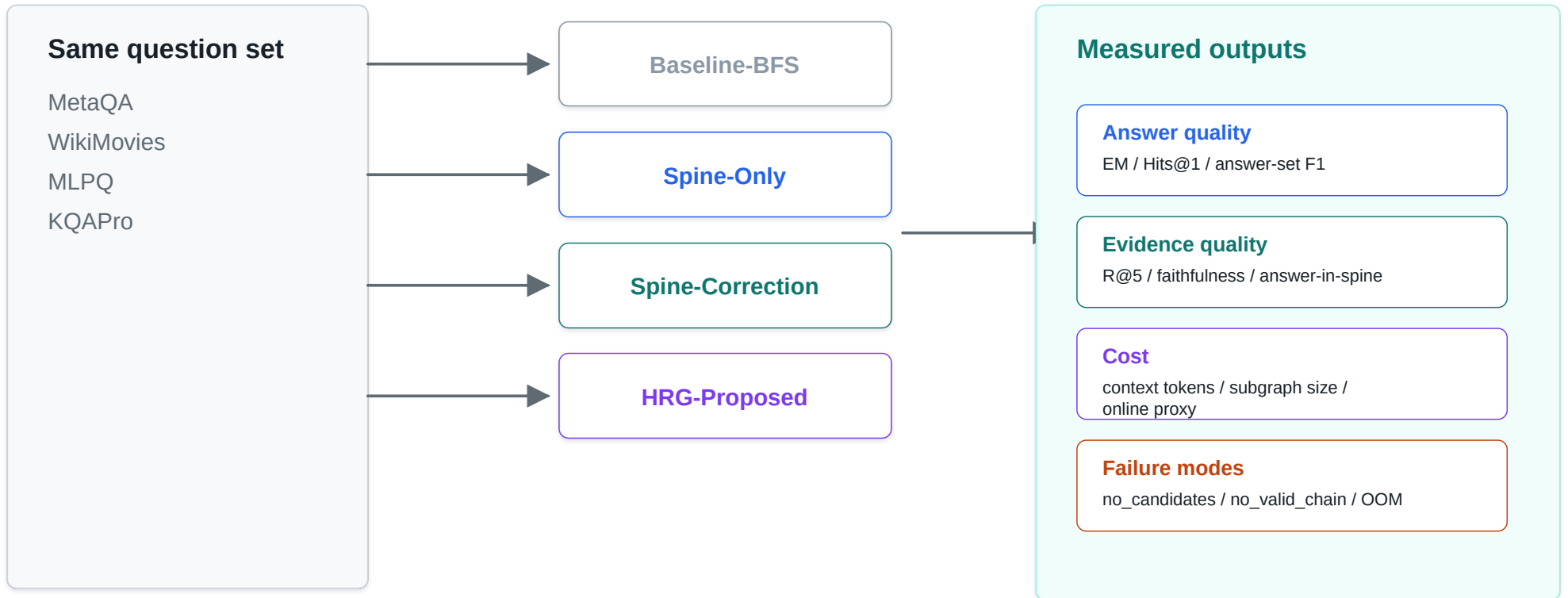
How many counties have population > 7800 or < 40000000?

Required structure

**operators: FindAll, FilterNum, Or, Count**

**HRG-Proposed is strongest when question semantics align with reusable KG relation structures.**

## Evaluation design: answer quality, evidence quality, and cost



**Do not rank by EM alone: defend the quality-cost-evidence trade-off.**